

Inputs

Clinical table

CSV, Excel or Parquet

Omics matrix

CSV, TSV, Parquet, or .gz files as GEO and Xena serve them; TCGA barcodes matched

External cohort

same or another platform

Checks run by default

Survival curves and Cox regression

Kaplan-Meier with numbers at risk, log-rank, Efron Cox model, proportional-hazards tests

Marker evaluation

score tests for added value over clinical data
→ family-wise error by permutation (Westfall-Young; marker residuals permuted)
→ whole screen repeated on subsamples: tiers
→ corrected C against clinical-only C

Prediction models

machine and deep learning on the same splits (holdout, repeated CV, locked test); paired intervals for the difference from Cox

Outputs

Figures and tables

forest plots, marker tables, leaderboards with intervals

Reporting checklists

REMARK and TRIPOD+AI filled from the run (Word, Markdown)

Locked model

hashed recipe applied unchanged to the external cohort

external validation, without refitting